

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours**Maximum Marks:35**

- Q.1** Write a C program to simulate Banker's algorithm for the purpose of Deadlock avoidance. The following snapshot of system, A, B, C and D are the resource type.

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

Implement the following Menu.

- a. Accept Available
- b. Display Allocation, Max
- c. Display the contents of need matrix
- d. Display Available [15 Marks]

- Q.2** Write the simulation program for demand paging and show the page scheduling and total number of page faults according the optimal page replacement algorithm. Assume the memory of n frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2 [15 Marks]

- Q.3.** Oral/Viva [05 Marks]